

	Since $V = V_3$, the rock would travel on into space.
3(a)(i)	Using the usual notation, for the ideal gas at point A, $pV_A = nRT_A$ $\rightarrow p(2.8 \times 10^3) = nR(315) \quad \text{---- (1)}$ For the ideal gas at point B, $pV_B = nRT_B$ $\rightarrow p(7.6 \times 10^3) = nRT_B \quad \text{---- (2)}$ $\frac{(2)}{(1)} : \quad \frac{7.6 \times 10^3}{2.8 \times 10^3} = \frac{T_B}{315} \rightarrow T_B = 855 \text{ K}$
3(a)(ii)	The external work done by the gas, $W = p \Delta V = p (V_B - V_A)$ $= (3.5 \times 10^5)[(7.6 - 2.8) \times 10^3] \times 10^{-6}$ $= 1680 \text{ J}$
3(b)	From B to C: there is no change in temperature, so there is no change in internal energy, i.e. $\Delta U_1 = 0$ From C to A: there is no change in volume, so there is no work done, i.e. $W = 0$. The change in internal energy, $\Delta U_2 = Q + W = Q = -2520 \text{ J}$ From A to B: let the change in internal energy be ΔU_3

No.	Solution								
	For a complete cycle, $B \rightarrow C \rightarrow A \rightarrow B$, the total change in internal energy = 0, i.e. $\Delta U_1 + \Delta U_2 + \Delta U_3 = 0 \rightarrow \Delta U_3 = 0 - \Delta U_1 - \Delta U_2$ $= 0 - 0 - (-2520)$ $= 2520 \text{ J}$ <table data-bbox="908 692 2193 956"> <tr> <th>stage</th><th>increase in internal energy / J</th></tr> <tr> <td>from point B to point C</td><td>0</td></tr> <tr> <td>from point C to point A</td><td>-2520</td></tr> <tr> <td>from point A to point B</td><td>2520</td></tr> </table>	stage	increase in internal energy / J	from point B to point C	0	from point C to point A	-2520	from point A to point B	2520
stage	increase in internal energy / J								
from point B to point C	0								
from point C to point A	-2520								
from point A to point B	2520								
4(a)(i)	At equilibrium								